

WORKED EXAMPLE

Keep memory when the mini-batch changes

Chosen small batches make the stored optimizer state visible.

$$\hat{y} = w, \quad \ell(w; t) = \frac{1}{2}(w - t)^2, \quad \frac{d\ell}{dw} = w - t$$

$$A = (1, 3), \quad B = (-3, -1), \quad w_0 = 0, \quad v_0 = 0$$

$$\eta = 0.1, \quad \beta = 0.5$$

w is the current parameter. v is the previous signed displacement: positive means the last update increased w . Beta retains half of that displacement.

Process A then B in epoch1, followed by A in epoch2. This explicitly fixed order supports hand calculations; lecture algorithms shuffle the data.

WORKED EXAMPLE

Derive each batch gradient before updating

Average the two example gradients at one common evaluation point.

$$g_A(w) = \frac{(w - 1) + (w - 3)}{2} = \frac{2w - 4}{2} = w - 2$$

$$g_B(w) = \frac{(w + 3) + (w + 1)}{2} = \frac{2w + 4}{2} = w + 2$$

A pulls toward its target mean 2; B pulls toward its target mean -2. The loss changes with the selected batch, even though the model and optimizer settings stay fixed.

Reset the temporary sum of example gradients at the start of each batch. Divide by its actual size 2 once. Do not reset w or v .

WORKED EXAMPLE

Map the lecture equations to our state

Lecture5 pages103,106 and109; v is the slide displacement ΔW .

$$\text{Momentum : } v_{new} = \beta v - \eta g_B(w), \quad w_{new} = w + v_{new}$$

$$\text{Nesterov : } q = w + \beta v, \quad v_{new} = \beta v - \eta g_B(q)$$

$$w_{new} = w + v_{new} = q - \eta g_B(q)$$

Here B means whichever batch is current. The lookahead q is temporary; every example of that batch uses this same q . It is not an extra stored training update.

The slide first moves in place to q , then applies the correction. Our saved-old- w form is algebraically identical. Do not add βv twice or insert an extra $(1-\beta)$ factor.

WORKED EXAMPLE

Both methods: first A gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0, \quad v = 0, \quad \text{targets} = (1, 3)$$

Both methods use w here: the initial displacement is zero, so Nesterov also has $q = w + 0.5(0) = w$.

$$g_1 = 0 - (1) = -1, \quad g_2 = 0 - (3) = -3$$

$$g = \frac{(-1) + (-3)}{2} = -2$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED EXAMPLE

Both methods: first A update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(0) - 0.1(-2) = 0.2$$

$$w_{new} = 0 + (0.2) = 0.2$$

$$(w, v) \text{ to retain} = (0.2, 0.2)$$

Save both new values for the next batch. Only the temporary gradient sum is discarded.

WORKED EXAMPLE

Momentum: B gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0.2, \quad v = 0.2, \quad \text{targets} = (-3, -1)$$

Ordinary momentum evaluates this batch at the stored w , before adding the retained displacement.

$$g_1 = 0.2 - (-3) = 3.2, \quad g_2 = 0.2 - (-1) = 1.2$$

$$g = \frac{(3.2) + (1.2)}{2} = 2.2$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED EXAMPLE

Momentum: B update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(0.2) - 0.1(2.2) = -0.12$$

$$w_{new} = 0.2 + (-0.12) = 0.08$$

$$(w, v) \text{ to retain} = (0.08, -0.12)$$

Save both new values for the next batch. Only the temporary gradient sum is discarded.

WORKED EXAMPLE

Nesterov: B gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0.2, \quad v = 0.2, \quad \text{targets} = (-3, -1)$$

$$q = 0.2 + 0.5(0.2) = 0.3$$

$$g_1 = 0.3 - (-3) = 3.3, \quad g_2 = 0.3 - (-1) = 1.3$$

$$g = \frac{(3.3) + (1.3)}{2} = 2.3$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED EXAMPLE

Nesterov: B update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(0.2) - 0.1(2.3) = -0.13$$

$$w_{new} = 0.2 + (-0.13) = 0.07$$

$$(w, v) \text{ to retain} = (0.07, -0.13)$$

For Nesterov this is also q minus the gradient correction. Save both new values; discard the temporary gradient sum and temporary lookahead.

WORKED EXAMPLE

Continue into a new epoch without resetting state

Each method carries its own state from B into the next A.

Momentum : $(w, v) = (0.08, -0.12)$

Nesterov : $(w, v) = (0.07, -0.13)$

An epoch boundary is a data-pass boundary. It does not restart optimizer memory. The next two calculations continue these states; they are not new comparisons from zero.

A new batch starts with a fresh gradient accumulator. A new training run initializes w and v once. Confusing these two resets changes the algorithm.

WORKED EXAMPLE

Momentum: next-epoch A gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0.08, \quad v = -0.12, \quad \text{targets} = (1, 3)$$

Ordinary momentum evaluates this batch at the stored w , before adding the retained displacement.

$$g_1 = 0.08 - (1) = -0.92, \quad g_2 = 0.08 - (3) = -2.92$$

$$g = \frac{(-0.92) + (-2.92)}{2} = -1.92$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED EXAMPLE

Momentum: next-epoch A update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(-0.12) - 0.1(-1.92) = 0.132$$

$$w_{new} = 0.08 + (0.132) = 0.212$$

$$(w, v) \text{ to retain} = (0.212, 0.132)$$

Save both new values for the next batch. Only the temporary gradient sum is discarded.

WORKED EXAMPLE

Nesterov: next-epoch A gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0.07, \quad v = -0.13, \quad \text{targets} = (1, 3)$$

$$q = 0.07 + 0.5(-0.13) = 0.005$$

$$g_1 = 0.005 - (1) = -0.995, \quad g_2 = 0.005 - (3) = -2.995$$

$$g = \frac{(-0.995) + (-2.995)}{2} = -1.995$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED EXAMPLE

Nesterov: next-epoch $\mathbf{A}$ update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(-0.13) - 0.1(-1.995) = 0.1345$$

$$w_{new} = 0.07 + (0.1345) = 0.2045$$

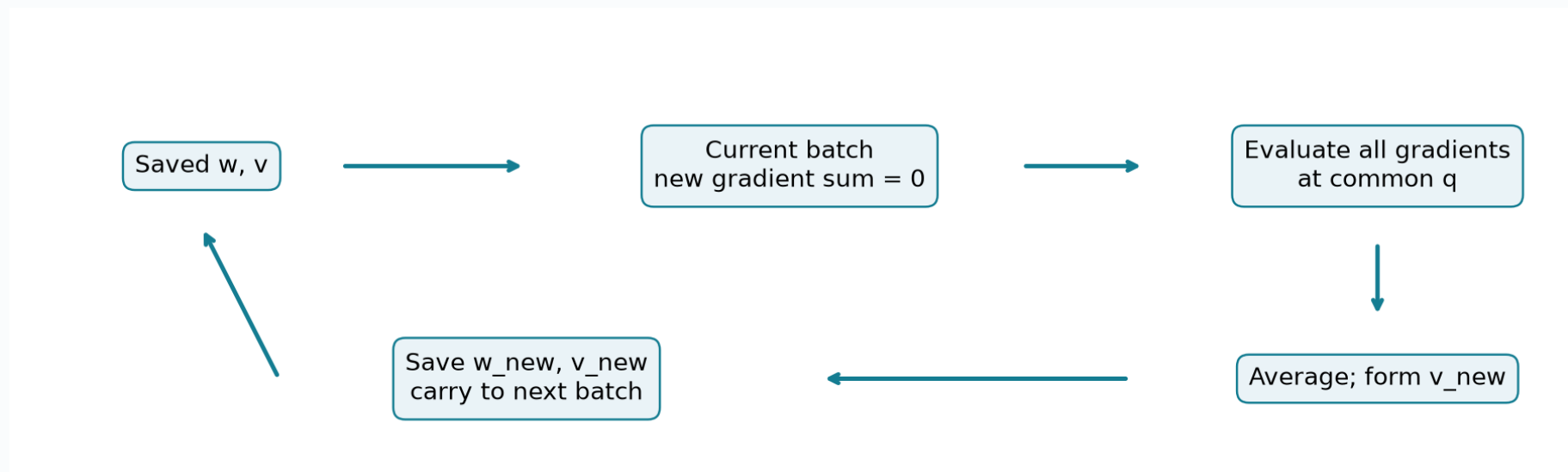
$$(w, v) \text{ to retain} = (0.2045, 0.1345)$$

For Nesterov this is also q minus the gradient correction. Save both new values; discard the temporary gradient sum and temporary lookahead.

WORKED EXAMPLE

Persistent state versus temporary batch quantities

Keep the saved state across batches and across epochs.



The flow applies to each batch. Momentum uses $q=w$; Nesterov uses $q=w+\text{beta}*v$. There is one final stored parameter update per batch.

WORKED EXAMPLE

Read the three-batch traces together

Each row starts from the preceding row of the same method.

method / batch	gradient point	mean gradient	new v	new w
Momentum / A	0	-2	0.2	0.2
Momentum / B	0.2	2.2	-0.12	0.08
Momentum / A again	0.08	-1.92	0.132	0.212
Nesterov / A	0	-2	0.2	0.2
Nesterov / B	0.3	2.3	-0.13	0.07
Nesterov / A again	0.005	-1.995	0.1345	0.2045

After B, negative displacement is retained. On the next A, the new batch gradient reverses the update again. Momentum smooths changing directions; it does not eliminate every reversal or guarantee a better loss.

INDEPENDENT PRACTICE

Your turn: change both mini-batches

Start a fresh run for each method; retain $\eta=.1$ and $\beta=.5$.

$$A = (0, 2), \quad B = (2, 4), \quad w_0 = v_0 = 0$$

Calculate A then B using ordinary momentum, then repeat from the same initial state using Nesterov. Show both example gradients, their average, each displacement and each parameter.

For Nesterov, explicitly calculate the common lookahead before both B gradients. Compare the correct B update with the mistaken procedure that resets v to zero just before B.

WORKED ANSWER

Both methods: practice A gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0, \quad v = 0, \quad \text{targets} = (0, 2)$$

Both methods use w here: the initial displacement is zero, so Nesterov also has $q = w + 0.5(0) = w$.

$$g_1 = 0 - (0) = 0, \quad g_2 = 0 - (2) = -2$$

$$g = \frac{(0) + (-2)}{2} = -1$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED ANSWER

Both methods: practice A update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(0) - 0.1(-1) = 0.1$$

$$w_{new} = 0 + (0.1) = 0.1$$

$$(w, v) \text{ to retain} = (0.1, 0.1)$$

Save both new values for the next batch. Only the temporary gradient sum is discarded.

WORKED ANSWER

Momentum: practice B gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0.1, \quad v = 0.1, \quad \text{targets} = (2, 4)$$

Ordinary momentum evaluates this batch at the stored w , before adding the retained displacement.

$$g_1 = 0.1 - (2) = -1.9, \quad g_2 = 0.1 - (4) = -3.9$$

$$g = \frac{(-1.9) + (-3.9)}{2} = -2.9$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED ANSWER

Momentum: practice B update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(0.1) - 0.1(-2.9) = 0.34$$

$$w_{new} = 0.1 + (0.34) = 0.44$$

$$(w, v) \text{ to retain} = (0.44, 0.34)$$

Save both new values for the next batch. Only the temporary gradient sum is discarded.

WORKED ANSWER

Nesterov: practice B gradient

Carry the parameter and displacement shown below into this batch.

$$w = 0.1, \quad v = 0.1, \quad \text{targets} = (2, 4)$$

$$q = 0.1 + 0.5(0.1) = 0.15$$

$$g_1 = 0.15 - (2) = -1.85, \quad g_2 = 0.15 - (4) = -3.85$$

$$g = \frac{(-1.85) + (-3.85)}{2} = -2.85$$

Both example gradients use the same evaluation point. Their average is the gradient supplied to this one optimizer update.

WORKED ANSWER

Nesterov: practice B update

Form a new displacement, then add it to the saved old parameter.

$$v_{new} = \beta v - \eta g$$

$$v_{new} = 0.5(0.1) - 0.1(-2.85) = 0.335$$

$$w_{new} = 0.1 + (0.335) = 0.435$$

$$(w, v) \text{ to retain} = (0.435, 0.335)$$

For Nesterov this is also q minus the gradient correction. Save both new values; discard the temporary gradient sum and temporary lookahead.

WORKED ANSWER

Answer: why resetting memory gives a different result

This comparison concerns the fresh practice batches, not the main example.

$$w_{Momentum} = 0.44, \quad w_{Nesterov} = 0.435$$

$$\text{wrong reset: } w = 0.1, \quad v = 0 \implies q = 0.1$$

$$g_B(0.1) = \frac{(0.1 - 2) + (0.1 - 4)}{2} = -2.9$$

$$v_{new} = 0.5(0) - 0.1(-2.9) = 0.29$$

$$w_{new} = 0.1 + 0.29 = 0.39$$

Both mistaken reset procedures give .39 here. The missing old displacement also removes the Nesterov lookahead shift. Keep optimizer memory; reset only batch-local accumulators.